MIL-L-80005A

18 May 1970

Superseding
MIL-L-80005
4 May 1964

MILITARY SPECIFICATION

LAPPING MACHINES, METALWORKING, HIGH PRECISION, * FLAT SURFACES, SINGLE FACE, FREE-ABRASIVE TYPE

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers single face, flat surface lapping machines having a metal lapping plate and utilizing free-abrasives suspended in a liquid vehicle as a lapping medium. Although primarily intended for lapping metal parts, the machines covered herein are also capable of lapping parts made of materials such as carbon, ceramics, plastics, quartz, glass, electronic semi-conductor materials, and other similar materials.

* 1.2 Classification. This specification covers machines of the following types and sizes. The type and size required for specific procurements shall be as specified (see 6.2).

Type I - Bench Mounted

Size 9 - 9 inch diameter lapping plate
Size 12 - 12 inch diameter lapping plate
Size 15 - 15 inch diameter lapping plate

Type II - Floor Mounted

Size 18 - 18 inch diameter lapping plate
Size 24 - 24 inch diameter lapping plate
Size 28 - 28 inch diameter lapping plate
Size 32 - 32 inch diameter lapping plate
Size 36 - 36 inch diameter lapping plate
Size 48 - 48 inch diameter lapping plate
Size 64 - 64 inch diameter lapping plate
Size 72 - 72 inch diameter lapping plate
Size 84 - 84 inch diameter lapping plate

FSC 3419

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2. APPLICABLE DOCUMENTS

* 2.1 Military publications. The following documents of the issue in effect on the date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MIL-M-18058	Machinery, Metal and Woodworking, Support Equipment and Associated Repair Parts, Preparation for Delivery of

STANDARDS

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS HANDBOOK H28

Screw-Thread Standards for Federal Services

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402.)

JIC ELECTRICAL STANDARDS FOR GENERAL PURPOSE MACHINE TOOLS

(Application for copies should be addressed to the National Machine Tool Builders' Association, 2139 Wisconsin Avenue, N. W., Washington, D. C. 20007.

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AMERICAN GEAR MANUFACTURERS ASSOCIATION AGMA 360.01

Manual for Machine Tool Gearing

(Application for copies should be addressed to the American Gear Manufacturers Association, One Thomas Circle, N. W., Washington, D. C. 20005.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.

3. REQUIREMENTS

* 3.1 First Article approval. If First Article approval is required (see 6.2 and 6.3), inspection therefor shall be in accordance with Section 4. A First Article comprises a preproduction model or initial production sample; approval involves testing and evaluating the First Article for conformance with the requirements herein before, or in the initial stage of, production under a contract.

* 3.2 Design. The machine shall be one of the manufacturer's current models, new and unused. The machine shall be capable of lapping parts in accordance with the requirements specified herein. The design and construction shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

* 3.2.1 Controls. All operating controls shall be located convenient to the operator.

* 3.2.2 Safety devices. The machine shall have covers or other safety devices for all parts that present safety hazards. The devices shall include guards for belts, pulleys, chains, gears, and sprockets. All guards shall provide easy access to the guarded parts, and shall not interfere with operation of the machine. When special safety devices are required, they shall be as specified (see 6.2).

* 3.2.3 Lubrication. Means shall be provided to insure adequate lubrication for all moving parts. Recirculating systems shall include filters which are cleanable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings, and filler caps shall be accessible.

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* 3.2.4 Interchangeability. To provide for replacement of worn parts, all parts produced by the same manufacturer and bearing the same part number shall be functionally interchangeable, and shall be dimensionally identical within the manufacturing tolerance limits in use by their manufacturer.

* 3.3 Construction. The machine shall be so constructed that when installed and connected to power, it shall be ready for operation. The machine shall be constructed of parts which are without defect and free of repair. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

* 3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale and mismatching. Processes such as welding, peening, plugging, or filling with solder or metallic paste shall not be used for reclaiming any defective part for use in the machine.

* 3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these operations shall be employed as a repair measure for any defective part.

* 3.3.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed in such a manner as to prevent change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

* 3.3.4 Surfaces. All surfaces of castings, forgings, molded parts, stampings and welded parts shall be cleaned and free from sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be commensurate with the manufacturer's standard commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2), the machine shall be painted in accordance with the manufacturer's standard commercial practice.

3.3.6 Threads. All threaded parts shall be in the inch system and shall conform to Handbook H28.

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* 3.3.7 Plates. All words on speed and feed indicating plates and on instruction plates shall be in the English language, engraved, etched, embossed or stamped in bold-face letters on a contrasting background.

* 3.3.8 Gears. All gears used in the machine and its components shall be machined in the inch system and shall conform to the applicable requirements of AGMA 360.01.

* 3.4 Components. The lapping machine shall consist essentially of a base, lapping plate, lapping plate drive system, workholder rings, workholder ring arms, abrasive system, pressure plate system, cycle timer, and other components described herein.

* 3.4.1 Base. The base shall have adequate strength to support all components and a full load of workpieces without permanent deformation. The base of Type I machines shall be designed for mounting onto a workbench or stand. The base of Type II machines shall be designed for mounting onto the floor or a foundation. The base height of Type II machines shall be such that the lapping plate is at a convenient height for loading and unloading the workpieces. Access openings shall be provided for servicing all components mounted within the base.

* 3.4.2 Machine stand. When specified (see 6.2), a machine stand shall be provided for Type I machines. The stand shall be a steel weldment capable of rigidly supporting the machine at a convenient operating height.

* 3.4.3 Lapping plate. Unless otherwise specified (see 6.2), the lapping plate shall be flat surfaced and shall be of either the solid type or the replaceable segment type. The lapping plate shall be constructed of a grade of cast iron or steel alloy entirely suitable for lapping plate use with silicon carbide, boron carbide, or aluminum oxide lapping slurries. The plate shall be stress relieved to minimize distortion and the surface finish and flatness of the working surface shall be such that the machine can meet all tests and performance requirements specified herein. Unless otherwise specified (see 6.2), the working surface of the plate shall be grooved to assist in distributing fresh slurry to the workpieces and discharging spent slurry and abraded metal particles. The grooves shall be of adequate depth to allow utilization of the useful working thickness of the plate. The plate shall be self-conditioning during use to maintain flatness and means shall be provided for adjusting the self-conditioning system to compensate for non-uniform wear of the plate working surface. The diameter of the lapping plate shall conform to Table I.

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* 3.4.4 Lapping plate drive system. The lapping plate shall be supported on a vertical spindle and shall be driven by an electric motor conforming to 3.7.1, and a reduction gear drive mechanism. The drive system shall drive the lapping plate at one or more speeds suitable for meeting the performance requirements stated herein. The reduction gear drive mechanism shall be completely enclosed and the housing shall be sealed to prevent loss of lubricant or entrance of abrasive slurry or other extraneous materials. The reduction gear housing shall have means for filling, draining, and observing the lubricant level. All revolving parts in the drive system shall be mounted on bearings of a size and type suitable for the application. All gears in the drive system shall conform to AGMA 360.01.

* 3.4.5 Workholder rings (plate conditioning rings). The machine shall have the quantity and size of workholder rings specified in Table I. The rings shall be designed to contain the parts to be lapped and also provide a continuous trueing action to the lapping plate during operation. The rings shall be positioned so that, during operation, the entire surface of the lapping plate is uniformly abraded by the rings and the workpieces contained therein. The rings shall revolve during operation either by frictional contact with the lapping plate or by means of a geared drive system. The ring design shall allow free passage of abrasive slurry to and from the workpieces loaded into the rings and shall allow free random movement of the workpieces. The rings shall have adequate weight to provide a constant trueing action to the lapping plate without producing excessive wear on either the lapping plate or the rings themselves. On machines not equipped with a pneumatically operated pressure plate system, each ring shall have an arm conforming to 3.4.6 to prevent the rings from rotating about the axis of the lapping plate. On machines equipped with a pneumatically operated pressure plate system (see 3.4.8), the pressure plates may prevent such rotation.

* 3.4.6 Workholder ring arms. The workholder ring arms shall be attached to a non-rotating part of the machine and shall have one or more rollers which contact the rings, allowing the rings to revolve freely about their own axis, but restraining them from rotating about the axis of the lapping plate. The rollers shall be mounted on ball or roller bearings which are sealed to prevent loss of lubricant or entrance of abrasive slurry or other extraneous materials. Means shall be provided to allow the workholder ring arms to be raised or lowered to compensate for wear of the workholder rings and lapping plate. On machines which do not reverse the direction of ring rotation for correction of irregular wear of the lapping plate, the arms shall have an adjustment for repositioning the rings to compensate for such irregular wear.

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* 3.4.7 Weighted pressure plate. Unless otherwise specified (see 6.2), a weighted pressure plate shall be provided for each ring for use when lapping parts that have insufficient weight to provide adequate lapping pressure. The pressure plates shall have an outside diameter not more than 3/16 inch smaller than the inside diameter of the workholder rings and shall have sufficient weight to exert adequate lapping pressure. Each pressure plate shall have an eyebolt, handle, or other lifting device to facilitate removal of the pressure plates from the rings.

3.4.8 Pneumatically operated pressure plate systems.

3.4.8.1 Pneumatic raising and lowering with pneumatically controlled lapping pressures (Type II machines only). When specified (see 6.2), Type II machines shall have a pneumatic system that will raise and lower the pressure plates to facilitate loading and unloading of the machine and also allow the lapping pressure to be pneumatically controlled. The system shall consist of an air cylinder and piston assembly, pressure regulating valve, control valve, pressure gauge and pressure plate for each workholder ring and necessary air filters, mist lubricators, piping, tubing, and hoses. The system shall be designed to operate on a shop air supply of 90 to 110 pounds per square inch and shall provide regulated lapping pressures ranging from 1, or less, to 6, or more, pounds per square inch. The control valve shall have not less than three operating positions, one for rapidly raising the pressure plates under full pneumatic pressure, one for "inching" the plate downward to facilitate positioning the workholder rings around the pressure plate, and one for applying a regulated pressure to the air cylinder.

3.4.8.2 Pneumatic raising and lowering with weight loaded pressure plates (sizes 18 and 28 machines only). When specified (see 6.2) sizes 18 and 28 machines shall have weight loaded pressure plates and a pneumatic system that will raise and lower the pressure plates to facilitate loading and unloading of the machine. The pneumatic system shall consist of one or more air cylinders and piston assemblies, a control valve, and necessary regulator valves, air filters, mist lubricators, piping and tubing. The pneumatic system shall be designed to operate on a 90 to 110 pounds per square inch shop air system. The control valves shall be designed to rapidly raise the pressure plates under full pneumatic pressure and allow the plates to be "inched" downward to facilitate positioning the workholder rings around the pressure plate.

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* 3.4.9 Abrasive slurry system. All machines except Type I, Size 9, shall have an abrasive slurry system capable of dispensing regulated quantities of silicon carbide, boron carbide, or aluminum oxide slurries. The system shall include a sump or tank, a motor driven pump and agitator, piping, control valves, discharge nozzles, and a spent-slurry collection device. The sump or tank shall be covered to prevent contamination of its contents, shall have means for draining and filling and shall have a capacity conforming to Table I. The pump shall be designed to pump abrasive slurries of the types specified above without damage or excessive wear. The agitator shall be capable of keeping free-abrasive particles, having a grit size of 90 to 800, uniformly suspended in the slurry vehicle. A collection trough around the lapping plate periphery shall collect spent slurry and direct it to a suitable waste container.

* 3.4.9.1 Ease of cleaning. To facilitate changeover from one type and grade of abrasive slurry to another, all parts of the abrasive slurry system shall have smooth internal surfaces, filleted or rounded corners and shall have no hidden pockets or obstructions which might capture abrasive particles and cause difficulties in cleaning the system.

* 3.4.10 Cycle timer. The machine shall have an electrically operated cycle timer which, when actuated, will automatically start the lapping plate and slurry pump and, after a preselected time interval, will automatically stop the plate and pump. Unless otherwise specified (see 6.2), the timer for Type I machines shall have a range of 0 to not less than 30 minutes and shall have half-minute graduations; timers for Type II machines shall have a range of 0 to not less than 60 minutes and shall have one minute graduations.

* 3.4.11 Parts handling table. When specified (see 6.2), Type II, Size 18, 24, 28, and 32 machines shall have a parts handling table to facilitate loading and unloading of the machine. The table shall be of the wrap-around type extending around the periphery of the lapping plate. The table shall have a working area not less than 50 percent greater than the total area of the workholder rings. The table shall have a vertical adjustment that will allow the table surface to be positioned level with the plane of the lapping plate.

* 3.5 Sizes and capacities. The sizes and capacities of the machine shall meet or exceed the requirements of Table I.

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* TABLE I

SIZES AND CAPACITIES

	TYPE I, BENCH MOUNTED			TYPE II, FLOOR MOUNTED								
	Size 9	Size 12	Size 15	Size 18	Size 24	Size 28	Size 32	Size 36	Size 48	Size 64	Size 72	Size 84
Lapping plate: Outside diameter, inches	9	12	15	18	24	28	32	36	48	64	72	84
Workholder rings: Quantity x inside diameter in inches	1 x 3-1/2	3 x 4-1/4 or 4 x 4	3 x 5-3/8	3 x 7-1/4 or 4 x 6-3/4	3 x 9-3/4 or 4 x 8-1/2	3 x 11-3/8 or 4 x 10	4 x 12	3 x 14-1/2 or 4 x 12-5/8	4 x 17	4 x 24	4 x 27-1/4	4 x 32
Abrasive slurry reservoir: Capacity, quarts	none	1/2	1-1/2	6	6	10	20	20	20	28	32	32
Motors: Lapping plate drive motor, horsepower	1/4	1/4	1/4	3/4	1	3	3	3	7-1/2	15	20	20
Abrasive slurry pump and agitator drive, horsepower	none	1/20	1/20	1/8	1/8	1/8	1/8	1/8	1/4	1/4	1/4	1/4

Note: Requirements are not less than those shown above.

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3.6 Performance accuracy.

* 3.6.1 Workpiece capacity. The machine shall be capable of lapping individual workpieces of such size that will fit into the workholder rings or as many small, similarly shaped and sized parts as can be loaded into the rings without crowding or interferring with free random movement of the pieces during the lapping process. Although intended operation is normally with the workholder rings installed on the lapping plate and with the workpieces loaded into the rings, the machine shall also be capable of lapping large parts, having a maximum dimension up to 60 percent of the diameter of the lapping plate, by removal of one or more of the workholder rings and employing special fixturing (not required to be furnished) for holding the part. During operation, with all workholder rings removed, the lapping plate is not required to be self-conditioning or self-trueing.

* 3.6.2 Performance and accuracy. The machine shall be capable of satisfactorily performing the tests specified in 4.5.2 and of meeting the surface flatness and surface finish requirements specified therein.

* 3.7 Electrical equipment. Unless otherwise specified (see 6.2), all electrical components and their locations shall conform to JIC Electrical Standards for General Purpose Machine Tools. An identified terminal shall be provided for grounding the machine when installed. The terminal shall be suitable for connection of the size grounding conductor specified in Table 15-1 of JIC Standards. Under-voltage protection shall be provided even though it may not be required by JIC Standards.

* 3.7.1 Motors. Motors shall have ball or roller bearings. Continuous duty horsepower ratings shall conform to the requirements of Table I. Motors of Type I machines shall operate on 115 volt, single phase, 60 hertz, alternating current. Motors of Type II machines shall be dual voltage 230/460 volt, 3 phase, 60 hertz, alternating current type initially wired for operation on the voltage specified (see 6.2). Unless otherwise specified (see 6.2), each motor shall be single speed and shall meet the requirements of a dripproof enclosure.

3.8 Standard equipment. Unless otherwise specified (see 6.2), the following equipment shall be furnished with each machine:

a. One set of hand tools normally furnished by the manufacturer.

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3.9 Optional equipment. Optional equipment and accessories shall be furnished as specified (see 6.2).

* 3.10 Electromagnetic interference control. Unless otherwise specified (see 6.2), equipment procured under this specification shall comply with the requirements of MIL-STD-461 for Class IIB equipment.

3.11 Fungus control. When required (see 6.2), fungus proofing shall be as specified.

* 3.12 Lubrication chart or plate. Unless otherwise specified (see 6.2), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed on a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means for mounting. The following information shall be furnished on the chart or plate:

Points of application

Service interval

Type of lubricant

Viscosity

Military or Federal Specification Number

* 3.13 Nameplate. Unless otherwise specified (see 6.2), a corrosion-resistant metal nameplate identifying the machine in accordance with MIL-STD-130 shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each such caption is clear as to its identity.

Nomenclature

Manufacturer's name

Manufacturer's model designation

Manufacturer's serial number

Power input (volts, total amps, phase, frequency)

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Diameter of lapping plate

Contract Number or Order Number

Federal Stock Number or Plant Equipment Code

Date of manufacture

U S

* 3.14 Identification plate. When specified (see 6.2), a separate identification plate of corrosion-resistant metal permanently and legibly marked in letters 3/16 inch high with the caption "Departmental or Activity Identification" shall be securely attached to each machine. The plate shall have blank space for the government to stamp 25 characters of the same size as the caption.

* 3.15 Technical data. Data shall be furnished as specified (see 6.2.2).

* 3.16 Workmanship. Workmanship of the machine and accessories shall be commensurate with the requirements specified herein and shall be of a quality prevailing among manufacturers normally producing equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsiblity for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification when such action is deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First Article inspection. When First Article approval is required under 3.1, First Article inspection shall be performed. Unless otherwise specified (see 6.2), First Article inspection shall comprise the examination in 4.4 and all tests in 4.5. Failure to pass the examination or any test shall be cause for disapproval of the First Article.

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* 4.3 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.4, the tests in 4.5.1 through 4.5.2 and the inspection in 4.6. Failure of an item to pass any examination or test shall be cause for rejection.

* 4.4 Examination. The machine and equipment shall be examined to determine compliance with the requirements in 3.2 through 3.5 and 3.7 through 3.16.

4.5 Tests

4.5.1 Operational test. The machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.5.2 Performance tests

* 4.5.2.1 Workpieces. The performance test shall be conducted using a sufficient quantity of disks to fill each workholder ring to not less than 60 percent of the calculated ring area (total disk contact area = 60 percent of total ring area, based on inside diameter). The disks shall be made of any grade of steel having a hardness not softer than Rockwell C-60. The disks shall have a surface finish not smoother than 20 microinches (arithmetic average), a diameter 15 to 25 percent of the inside diameter of the workholder rings and a random thickness variation not less than 0.004 inches. The thickness of the disks shall be at the manufacturer's option. The manufacturer may utilize any parts on hand meeting the above requirements or may cut the disks from suitable round bar stock and prepare them to conform to the above requirements.

* 4.5.2.2 Flatness and surface finish test. The workholder rings shall be filled to 60 percent of their area with the workpieces described in 4.5.2.1 and one surface of the workpieces shall be lapped. The lapping pressure, type and grade of abrasive and vehicle, and the lapping method shall be at the manufacturer's option. A roughing and finishing lapping operation may be performed, and a resilient pad may be used between the pressure plate and the workpieces to compensate for workpiece thickness variation. At the end of a 45 minute lapping cycle, the machine shall be stopped and 30 percent of the workpieces shall be randomly selected for measurement of surface finish and flatness. The surface finish of the lapped parts shall be not rougher than 2 microinches. Prior to the flatness measurement, the lapped surfaces may be hand polished only to the extent necessary to measure the flatness with an optical flat. The

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lapped surfaces shall be flat within ½ lightband per inch of workpiece diameter.

* 4.5.3 Electromagnetic interference control tests. Unless otherwise specified (see 6.2), equipment requiring electromagnetic interference control testing shall be tested for conformance with 3.10 in accordance with Table II of MIL-STD-461.

* 4.6 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine compliance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be in accordance with MIL-M-18058, at Level A or C as specified (see 6.2).

5.2 Packing. Packing shall be in accordance with MIL-M-18058, at Level A, B, or C as specified (see 6.2).

5.3 Marking. Marking shall be in accordance with MIL-STD-129. Additional marking shall be as specified (see 6.2).

6. NOTES

* 6.1 Intended use. The machines covered by this specification are primarily intended for lapping metal parts to a precise surface finish and surface flatness. The machines are also intended for lapping parts made of materials such as carbon, ceramics, plastic, quartz, glass, electronic semiconductor materials, and other similar materials.

* 6.2 Ordering data. Purchasers should exercise any desired option offered herein and procurement documents should specify the following:

* 6.2.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Type and size machine required (see 1.2).

c. First Article approval, if required (see 3.1 and 6.3).

d. Special safety devices, if required (see 3.2.2).

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e. Painting, if different (see 3.3.5).

f. If machine stand shall be provided with Type I machines (see 3.4.2).

g. Lapping plate if other than flat surfaced, solid or replaceable segment type (see 3.4.3).

h. If a non-grooved lapping plate shall be furnished (see 3.4.3).

i. Weighted pressure plates, if not required (see 3.4.7).

j. Type of pneumatically operated pressure plate system, if required (see 3.4.8.1 and 3.4.8.2).

k. Cycle timer range and graduations, if different (see 3.4.10).

l. Parts handling table, if required (see 3.4.11).

m. Electrical equipment, if different (see 3.7).

n. Voltage for initial wiring of motor (see 3.7.1).

o. Number of speeds and type of enclosure, if different (see 3.7.1).

p. Standard equipment, if different (see 3.8).

q. Optional equipment, if required; completely describe (see 3.9).

r. Electromagnetic interference control, if different (see 3.10 and 6.4).

s. Type fungus proofing, if required (see 3.11).

t. Lubrication chart, if different (see 3.12).

u. Nameplate, if different (see 3.13).

v. Identification plate, if different (see 3.14).

w. First Article inspection, if different (see 4.2).

x. Quality conformance inspection, if different (see 4.3).

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y. Electromagnetic interference control test requirements, if different (see 4.5.3 and 6.4).

z. Required level of preservation and packaging (see 5.1).

aa. Required level of packing (see 5.2).

bb. Additional marking, if required (see 5.3).

* 6.2.2 Contract data requirements. Required technical data, such as operator's manuals, parts lists, foundation and anchor bolt plans, wiring diagrams, and other instructions for operation and maintenance as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

* 6.3 First Article. When First Article approval is required under 3.1, the solicitation should include complete information concerning examination, test, and approval.

* 6.4 Army and Air Force procurements. Equipment purchased for the Departments of the Army and Air Force should have electromagnetic interference control tests in conformance with MIL-STD-461, Table II, Class IIB equipment, except that only Test Methods CE03 and RE02 should be required.

* 6.5 Marginal notations. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - WC
Navy - SH
Air Force - 84

Review Activities:
Army - WC
Navy - SH
Air Force - none

User Activities:
Army - SM, MU
Navy - MC, YD, AS, OS
Air Force - none

Preparing Activity:
DSA-IP

Project Number:
3419-0164

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